

Kappahl Enterprise Glossary Extraction Pipeline (Updated)

Overview

This document describes an enterprise-grade glossary extraction pipeline tailored for Kappahl-style fashion and PIM content. The pipeline focuses on extracting reusable multilingual terminology from localized product data using spaCy-based extraction rules, LaBSE semantic alignment, reviewer governance, and publishing approved glossary entries into an existing glossary storage application.

Updated Pipeline Scope

Kappahl PIM Content → Locale & Field Pairing → Fashion/PIM Text Normalization → Source + Target Term Extraction → LaBSE Semantic Alignment → Scoring & Filtering → Reviewer UI → Approved / Rejected / Edited Glossary Terms → Save / Publish to Existing Glossary App

Input Content

The pipeline should ingest localized PIM fields including:

- ProductName
- ProductNameShort
- Description
- ShortDescription
- LongDescription
- BulletPoints
- Material
- Composition
- Color
- Category
- Attribute values
- SEO fields

Long descriptions are used as mining sources for terminology extraction but should not become full glossary sentence pairs.

Text Normalization

Normalize source and target text before extraction:

- remove HTML

- decode entities
- normalize whitespace
- standardize punctuation
- normalize hyphenation
- preserve original casing

Examples:

- T shirt → T-shirt
- off white → off-white
- wide leg → wide-leg
- v neck → V-neck

Recommended Technology Stack

Primary Extraction:

- spaCy noun chunk extraction
- spaCy Matcher
- spaCy PhraseMatcher

Fallback Discovery:

- KeyBERT
- YAKE

Semantic Alignment:

- sentence-transformers/LaBSE

Storage & Vector Search:

- PostgreSQL + pgvector
- or Qdrant

Kappahl Domain Extraction Rules

High-priority extraction domains include:

Product Types

hoodie, cardigan, leggings, sweatshirt, bodysuit, jeans, blouse, jacket, dress

Materials

organic cotton, brushed fleece, ribbed jersey, recycled polyester, woven fabric

Sustainability Terms

organic cotton, certified cotton, recycled material, responsibly sourced cotton

Fit & Silhouette

relaxed fit, oversized fit, slim fit, wide leg, straight leg, high waist

Garment Construction

ribbed cuffs, drawstring waist, adjustable straps, zip closure, elastic waistband

Patterns & Appearance

floral print, holographic effect, washed look, embroidered detail

Baby & Kidswear

wrap bodysuit, padded snowsuit, anti-slip socks

Underwear & Nightwear

wireless bra, seamless underwear, pajama set

Extraction Rules

Suggested extraction patterns:

- ADJ + NOUN → organic cotton
- ADJ + ADJ + NOUN → soft brushed fleece
- MATERIAL + PRODUCT_TYPE → denim jacket
- FIT_ADJ + fit → relaxed fit
- LEG_SHAPE + leg → wide leg
- closure_type + closure → zip closure

Filtering Rules

Reject candidates that are:

- full sentences
- longer than 6 words
- generic marketing phrases
- punctuation-heavy
- one-off phrases
- SKU-like values

Blacklist examples:

- perfect for
- easy to wear
- comes with
- available in
- a great choice

LaBSE Semantic Alignment

Use LaBSE embeddings to align extracted source phrases with target-language phrases.

Alignment priority:

1. same product + same field
2. same product + any field
3. same category
4. global target phrase index

Recommended thresholds:

- ≥ 0.85 : strong candidate
- $0.75\text{--}0.85$: review candidate
- < 0.65 : reject

Scoring Model

final_score =

$$\begin{aligned} &0.30 \times \text{LaBSE_similarity} + \\ &0.20 \times \text{frequency_score} + \\ &0.15 \times \text{field_weight} + \\ &0.15 \times \text{domain_priority} + \\ &0.10 \times \text{translation_consistency} + \\ &0.10 \times \text{brand_relevance} \end{aligned}$$

Reviewer UI

The reviewer interface should support:

- approve candidates
- reject candidates
- edit source term
- edit target term
- merge duplicates

- mark preferred translations
- mark forbidden translations
- add usage notes
- bulk approve high-confidence terms
- filter by locale, category, domain, and confidence

Each candidate should display:

- source term
- target term
- source locale
- target locale
- category
- field origin
- frequency
- LaBSE score
- source context
- target context
- product evidence

Review Statuses

- needs_review
- approved
- rejected
- edited
- duplicate
- forbidden
- published

Glossary Candidate Schema

Suggested candidate structure:

- source_locale
- target_locale
- source_term
- target_term
- normalized_source_term
- normalized_target_term
- domain

- category
- field_origin
- frequency
- labse_score
- final_score
- status
- source_context
- target_context
- evidence_product_ids
- created_at
- updated_at

Publishing to Existing Glossary App

Publishing flow:

reviewed approved candidates → schema mapping → duplicate check → upsert into glossary app → mark as published

Recommended upsert key:

- source_locale
- target_locale
- normalized_source_term
- domain

Publishing metadata:

- published_at
- published_by
- external_glossary_entry_id
- publish_status
- publish_error

Enterprise Architecture

PIM Export / Event Stream → Content Ingestion Service → Normalization Service → spaCy Extraction Service → LaBSE Embedding Service → Vector Index → Alignment & Scoring Service → Candidate Store → Reviewer UI → Publishing Service → Existing Glossary App

Key Principle

Descriptions are mining sources.

Glossary entries are controlled reusable terminology.

Prioritize:

- material terminology
- sustainability vocabulary
- fit and silhouette terminology
- garment construction terms
- product-type consistency

Only short, reusable, brand-relevant terms should become approved glossary entries.